
Lorenzo Bini

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Education

- University of Geneva**, Ph.D. Candidate - Department of Computer Science & CUI Nov 2022 – Ongoing
- Major: Artificial Intelligence and Machine Learning.
 - Research interests: Graph neural networks, adversarial learning, representation learning, active and self-supervised learning. Generative AI for medicine and healthcare, including 3D genomics, multi-modal learning, and multi-omics/scRNA-seq data generation. Implementation of robust adversarial models within weak/self-supervised training strategies to reduce label acquisition costs.
- Polytechnic of Turin**, Master of Science in Physics of Complex Systems Sep 2020 – Sep 2022
- GPA: 4.0/4.0
- Alma Mater Studiorum - University of Bologna**, Bachelor degree in Physics & Astronomy Systems Sep 2017 – Sep 2020
- GPA: 4.0/4.0

Experience

- PhD Research Internship**, [Jian Ma's Lab](#) - Comp. Science Dept., Carnegie Mellon University - Joint collaboration with [Harvard University](#) and [Massachusetts Institute of Technology](#) Sep 2025 - Ongoing
- Developed and implemented generative models for epigenomics and gene regulation.
 - Built multi-modal learning models for multi-omics ensembles generation.
 - Designed and integrated LLM-based multi-modal encoders to fuse epigenomic signals and imaging data.
 - Implemented scalable training & inference (subchain batching, sparse graphs, RMSD alignment) and evaluation tools (contact-map, SCC, insulation score, radius-of-gyration).
 - Optimized memory and runtime (sparse attention) to enable large-region generation.
- PhD Researcher**, Hôpitaux Universitaires de Genève (HUG) – Geneva, Switzerland Nov 2022 - Ongoing
- Detection of minimal residual disease (MRD) of acute lymphoblastic and myeloid leukemia (AML/ALL) from flow cytometry data.
 - Development of fast training/inference deep learning models (e.g, Graph Transformers) for single-cell classification in weak/self-supervised contexts.
 - Development of multi-modal generative models (e.g., DDPMs, Flow Matching) for 3D genomics, flow cytometry, scRNA-seq, spatial transcriptomics, and multiomics data.
 - LLMs to streamline routine hospital processes.
- Teaching Assistant**, University of Geneva – Department of Computer Science & CUI Nov 2022 – Ongoing
- Introduction to Computational Finance - [14X030](#);
 - TALN: Traitement de la langue approches linguistiques et approches empiriques (NLP) - [34C2161](#);
 - Information Retrieval - [14X060](#);
 - Data Science - Analyse et Traitement de l'Information - [14X026](#);
 - Selected Chapters - Game Theory - [14X060](#);
- Research Assistant - Intern**, [Quantum Technology Group](#) - University of Norway Feb 2022 - July 2022
- Worked on quantum theory for entanglement and non-locality in optomechanics continuous variable systems.
 - Developed quantum frameworks to analyze two-cavity optomechanics systems.

Publications

- ChromWeave: Equivariant Generative Modeling of 3D Genome Organization** Apr 2026
Shuming Liu, *Lorenzo Bini*, Hongwei Tu, Jian Ma
[NeurIPS 2025 MLSB Workshop \(work done while at Carnegie Mellon University - Broad Institute\). Under double-blind review as a journal paper.](#)
- Probing Graph Neural Network Activation Patterns Through Graph Topology** Mar 2026
Florian Tori*, *Lorenzo Bini**, Marco Sorbi, Stéphane Marchand-Maillet, Vincent Ginis
[To appear in 2026 \(preprint, under double-blind review as a conference paper\).](#)
- LapDDPM: A Conditional Graph Diffusion Model for scRNA-seq Generation with Spectral Adversarial Perturbations** Jun 2025
Lorenzo Bini, Stéphane Marchand-Maillet
[Conference on Health, Inference, and Learning \(CHIL'26\), Seattle, WA.](#)
[ICML'2025 GenBio Workshop: The Second Workshop on Generative AI and Biology, Vancouver.](#)
- Self-Supervised Graph Learning via Spectral Bootstrapping and Laplacian-Based Augmentations** May 2025
Lorenzo Bini, Stéphane Marchand-Maillet
[Preprint. To appear in 2026, under double-blind review as a journal paper.](#)
- Massive Activations in Graph Neural Networks: Decoding Attention for Domain-Dependent Interpretability** Oct 2024
Lorenzo Bini, Marco Sorbi, Stéphane Marchand-Maillet
[ECAI'2025, Bologna \(oral presentation\) + ICLR'2025 Workshop XAI4Science: From Understanding Model Behavior to Discovering New Scientific Knowledge, Singapore.](#)
- Injecting Hierarchical Biological Priors into Graph Neural Networks for Flow Cytometry Prediction** Jul 2024
Lorenzo Bini, Stéphane Marchand-Maillet
[ICML'2024 + Workshop on Accessible and Efficient Foundation Models for Biological Discovery, Wien, Austria.](#)
- FlowCyt: A Comparative Study of Deep Learning Approaches for Multi-Class Classification in Flow Cytometry Benchmarking** Jun 2024
Lorenzo Bini, Margarita Liarou, Thomas Matthes, Stéphane Marchand-Maillet
[Conference on Health, Inference, and Learning \(CHIL'24\), New-York, NY.](#)
- Why Attention Graphs Are All We Need: Pioneering Hierarchical Classification of Hematologic Cell Populations with LeukoGraph** Feb 2024
Lorenzo Bini, Thomas Matthes, Stéphane Marchand-Maillet
[Preprint arXiv:2402.18610, under double-blind review as a conference paper.](#)
- HemaGraph: Breaking Barriers in Hematologic Single Cell Classification with Graph Attention** Dec 2023
Lorenzo Bini, Thomas Matthes, Stéphane Marchand-Maillet
[Preprint arXiv:2402.18611, under double-blind review as a conference paper.](#)

Awards & Oral Presentations

- Oral Presentation at the Machine Learning Summer School NYC hosted by Columbia University & Bloomberg - MLSS'26, New York** Jun. 2026
 - Invited oral presentation to expose my PhD research at the [Machine Learning Summer School NYC](#) hosted by [Columbia University](#) in collaboration with [Bloomberg](#) (MLSS'26), New York.
- Oral Presentation at the Second Workshop on Explainable Artificial Intelligence for the Medical Domain - ECAI'25, Bologna** Oct. 2025

- Invited oral presentation of the accepted paper "Massive Activations in Graph Neural Networks: Decoding Attention for Domain-Dependent Interpretability" at [European Conference on Artificial Intelligence \(ECAI'25\)](#), Bologna.

Winner of Swiss National Science Foundation PhD Mobility Grant Sep. 2025

- Winner of the Swiss National Science Foundation ([SNSF](#)) PhD Mobility Grant for a 6-months internship at Ray and Stephanie Lane Computational Biology Department - Carnegie Mellon University ([CMU](#)), Pittsburgh USA. Joint collaboration with Harvard University ([Broad Institute](#)) and Massachusetts Institute of Technology ([MIT EECS](#)).

PhD Symposium - CHIL'24 at Cornell Tech University, NY Jun. 2024

- Winner of the PhD Symposium money-prize to attend and present PhD work "Adversarial Robust GNNs: Enhancing Learning with Knowledge Injection in Single-Cell Data" at [CHIL'24](#), conference held by Cornell Tech University, New York.

CHAIR Structured Learning Workshop - Chalmers University of Technology Oct. 2023

- Oral presentation of the "Knowledge Distillation in Acute Myeloid Leukemia Classification: Tabular Data Meets Graph Neural Networks" poster at the [AI Structured Learning 2023 Workshop](#) in Göteborg, Sweden.

Winner of Thesis on Proposal 2021/2022 Feb. 2022

- Winner of the "Thesis on Proposal 2021/2022" call for bids for Master's Thesis on "Entanglement and nonlocality in optomechanics continuous variable systems" under the supervision of Prof. Francesco Pietro Massel & Prof. Vittorio Penna.
- Received grants for research period at USN-Kongsberg.

Projects

Flow Cytometry Deep Learning Benchmark [FlowCyt-Benchmark](#)

- Developed the first publicly available deep learning benchmark for single cell classification and clustering on flow cytometry data. Tested on a cohort of 30 patients selected by expert hematologists, from bone marrow and peripheral blood samples. Benchmarked multi-modal SOTA classification/generative models including GNNs, GraphTransformers, Diffusion Models (DDPMs), Flow Matching Models and VAEs.
- Tools Used: Python, CSS, HTML.

Hackathons & Competitions 2019 - Ongoing

- Regular participation at Kaggle/LeetCode competitions and Hackathons; runner-up BR41N.IO 2021 Hackathon@PoliTO, runner-up UNIBO-IBM-Unipol Hackathon 2019, 3rd classified QuHack4IA 2023.
- Tools Used: Python - Competitive Programming.

Math/Physics Olympiad Sep 2014 - Sep 2019

- 2x winner of the Italian Regional Math Olympiad.
- 1x winner of the Italian Regional Physics Olympiad.

Visiting Student, City Montessori School, Lucknow - Uttar Pradesh, India USN Aug 2016 - Sep 2016

- Visited the City Montessori School together with the italian cultural association "CinemáCinemá" to provide help and needs to elementary/mid school students.

Oxfam Volunteering 2016 - Ongoing

- I do regularly serve as volunteer for charity organization, such as [Oxfam Italy](#).

Technologies

- **Programming Languages:** Python, Julia, R, , Matlab, Mathematica, C/C++ , CUDA.
- **Frameworks:** Pytorch, Pytorch-Lightning, Tensorflow, Keras, Jupyter.
- **AI/ML:** SciPy, scikit-learn, Github, GitLab, Numpy, Pandas, Matplotlib, Seaborn, wandb, PyG.
- **Tools:** Git, Linux, ssh, SLURM, pip, Anaconda, Docker, VSCode, Bash (Shell).

Software Licenses: Kaluza Analysis Software, Zemax OpticStudio | Comprehensive Optical Design Software, TeXstudio, BioVinci Software.

Reviewing Duties

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| • Conference on Neural Information Processing Systems (NeurIPS). | 2025 - Ongoing |
| • International Conference on Learning Representations (ICLR). | 2023 - Ongoing |
| • International Conference on Machine Learning (ICML). | 2022 - Ongoing |
| • Learning on Graphs Conference (LOG). | 2023 - Ongoing |
| • AAAI Conference on Artificial Intelligence. | 2025 - Ongoing |
| • IEEE Transactions on Computational Biology and Bioinformatics. | 2023 - Ongoing |